STAT340 - Topic 1: Probability

Concepts

- Probability theory
- Random variables
- Independence
- Conditional probability
- Bayes' rule
- Distributions (PMF/PDF, CDF, inverse CDF)
- Expectation & variance
- Law of large numbers
- List of 340 RVs
- Covariance, correlation

Probability review

Suppose you have some **random process**, e.g. an experiment.

- An **outcome** is something that can be observed to happen,
- The set of **all possible outcomes** is denoted Ω
- Subsets $E \subseteq \Omega$, consisting of zero or more outcomes, are called **events**

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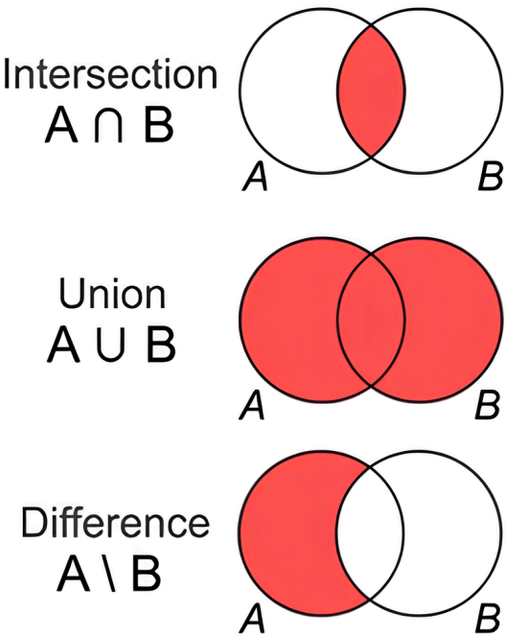
Examples:

- For flipping coins, $\Omega = \{H, T\}$, event of "heads" $= \{H\}$, event of "tails" $= \{T\}$
- For rolling 6-sided die, $\Omega = \{1, 2, 3, 4, 5, 6\}$. Some examples of events:
  - event: "getting a 6" $= \{6\}$
  - event: "getting no more than 3" $= \{1, 2, 3\}$, etc.
- For measuring physical dimensions, e.g. masses or lengths, $\Omega = [0, \infty)$

Event operations

For two events $A, B \subseteq \Omega$,

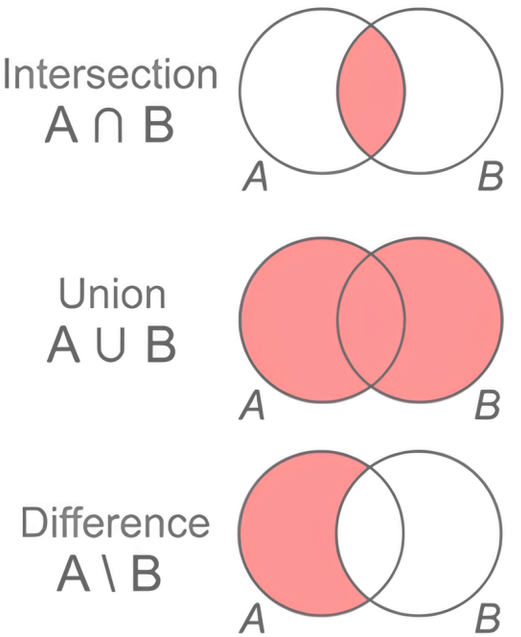
- The **intersection** $A \cap B$ is the set of all outcomes **in both** A, B, i.e. their overlap
  - If intersection is empty, they are **disjoint** or **mutually exclusive**
- The **union** $A \cup B$ is the set of all outcomes **in either** A, B or both, i.e. combining them
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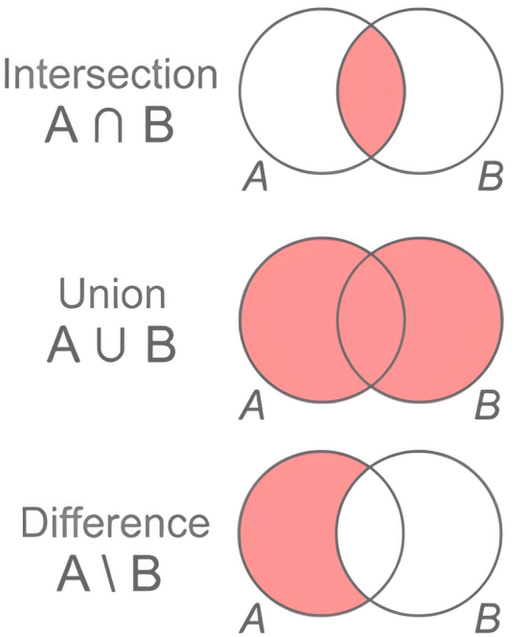
Exercise: roll a 6-sided die; let $A =$ "even", $B =$ "at least 3"

- $\Omega = ?$
- $A = ?$
- $B = ?$
- $A \cup B = ?$
- $A \cap B = ?$
- $A \backslash B = ?$
- $B \backslash A = ?$
- $\Omega \backslash (A \cup B) = ?$

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- $\Omega = \{1, 2, 3, 4, 5, 6\}$
- $A = \{2, 4, 6\}$
- $B = \{3, 4, 5, 6\}$
- $A \cup B = \{2, 3, 4, 5, 6\}$
- $A \cap B = \{4, 6\}$
- $A \backslash B = \{2\}$
- $B \backslash A = \{3, 5\}$
- $\Omega \backslash (A \cup B) = \{1\}$

Venn in the wild



source: http://redd.it/1n5dulo/

Probability function

A **probability function** $\Pr$ assigns real numbers to events, following these axioms:

1. Probability is nonnegative, i.e. $\Pr[E] \geq 0$ for any event E
2. Probability of all outcomes is 1, i.e. $\Pr[\Omega] = 1$
3. Probability of union of disjoint events is the sum of probability of each,
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Together, these imply the following corollaries:

1. Probability is always between $[0, 1]$
2. Probability of the **complement** (opposite) event $\Pr[E'] = 1 - \Pr[E]$
3. In general, for any events E_1, E_2, probability of their union:
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Exercise: suppose in CityTown, 70% of residents have a car, 40% have a bike, and 20% have both. How many have either a car OR a bike? How many have neither?

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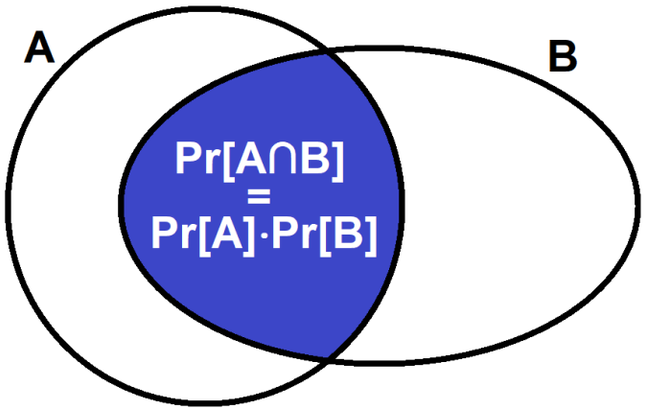
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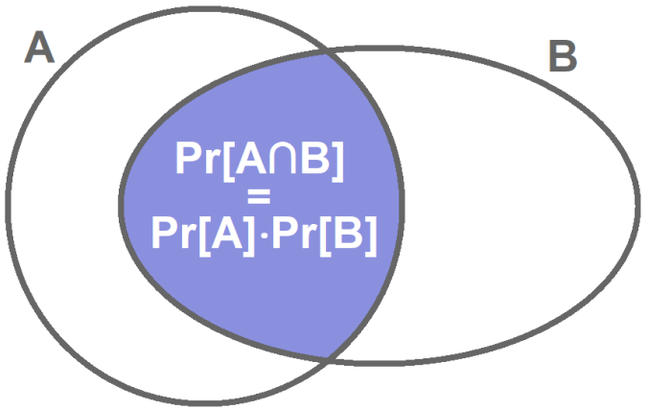


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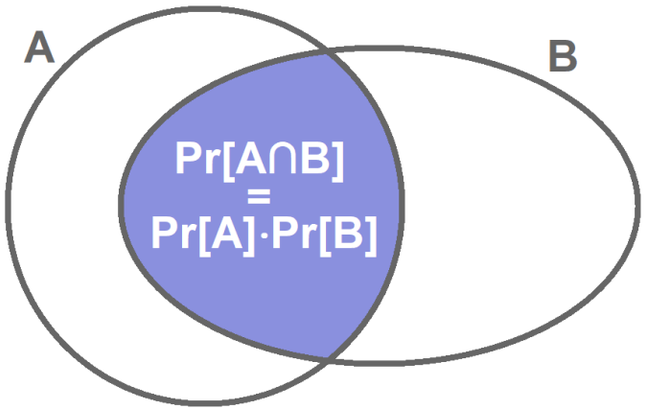


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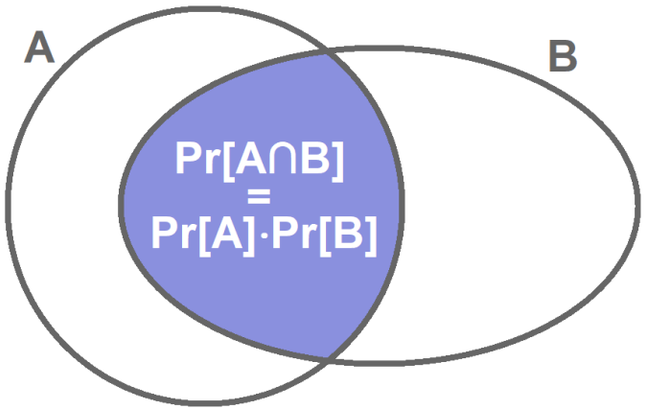
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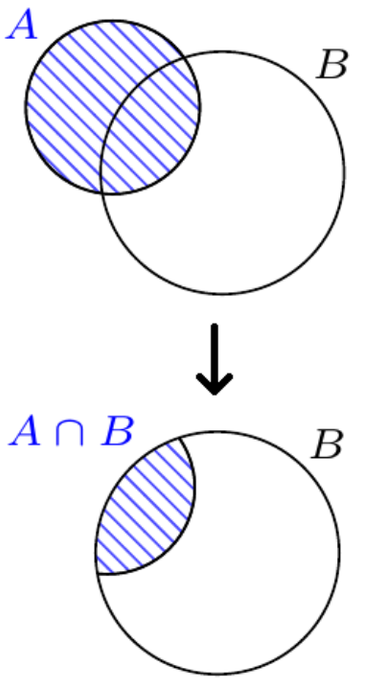
A, C dependent since $\Pr[A \cap C] = \frac{14}{60} = \frac{7}{30} \neq \frac{2}{3} \cdot \frac{1}{3} = \frac{40}{60} \cdot \frac{20}{60} = \Pr[A] \cdot \Pr[C]$

Exercise: Are B, C independent? If not, is there a better phrase to describe them?

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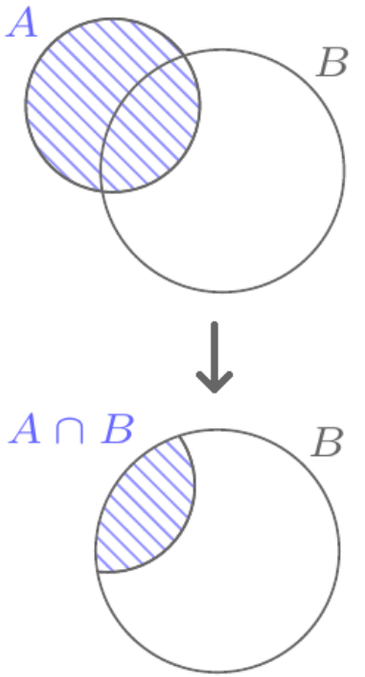
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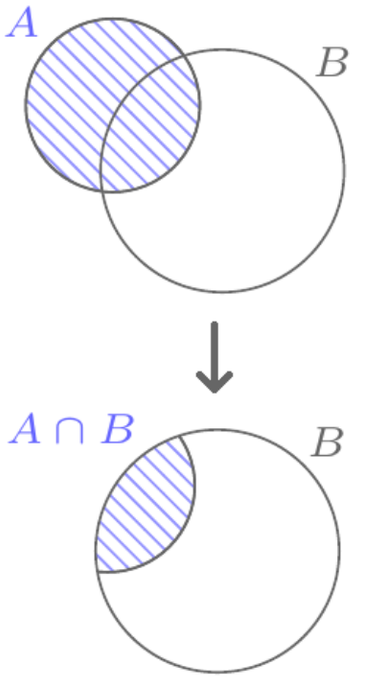
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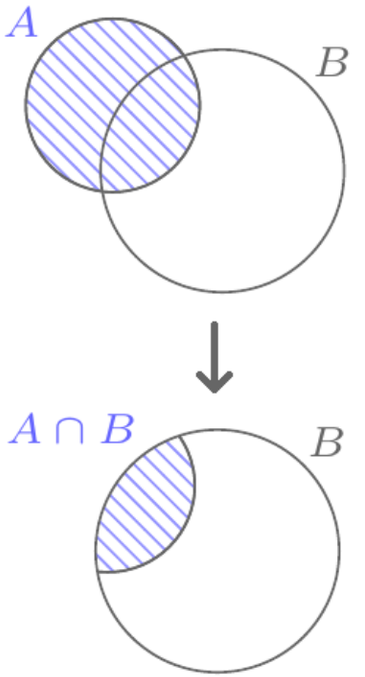
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Also note: $1 - \Pr[A \mid B] = \dfrac{\Pr[B]}{\Pr[B]} - \dfrac{\Pr[A \cap B]}{\Pr[B]} = \dfrac{\Pr[A' \cap B]}{\Pr[B]} = \Pr[A' \mid B]$

Conditional probability (cont.)

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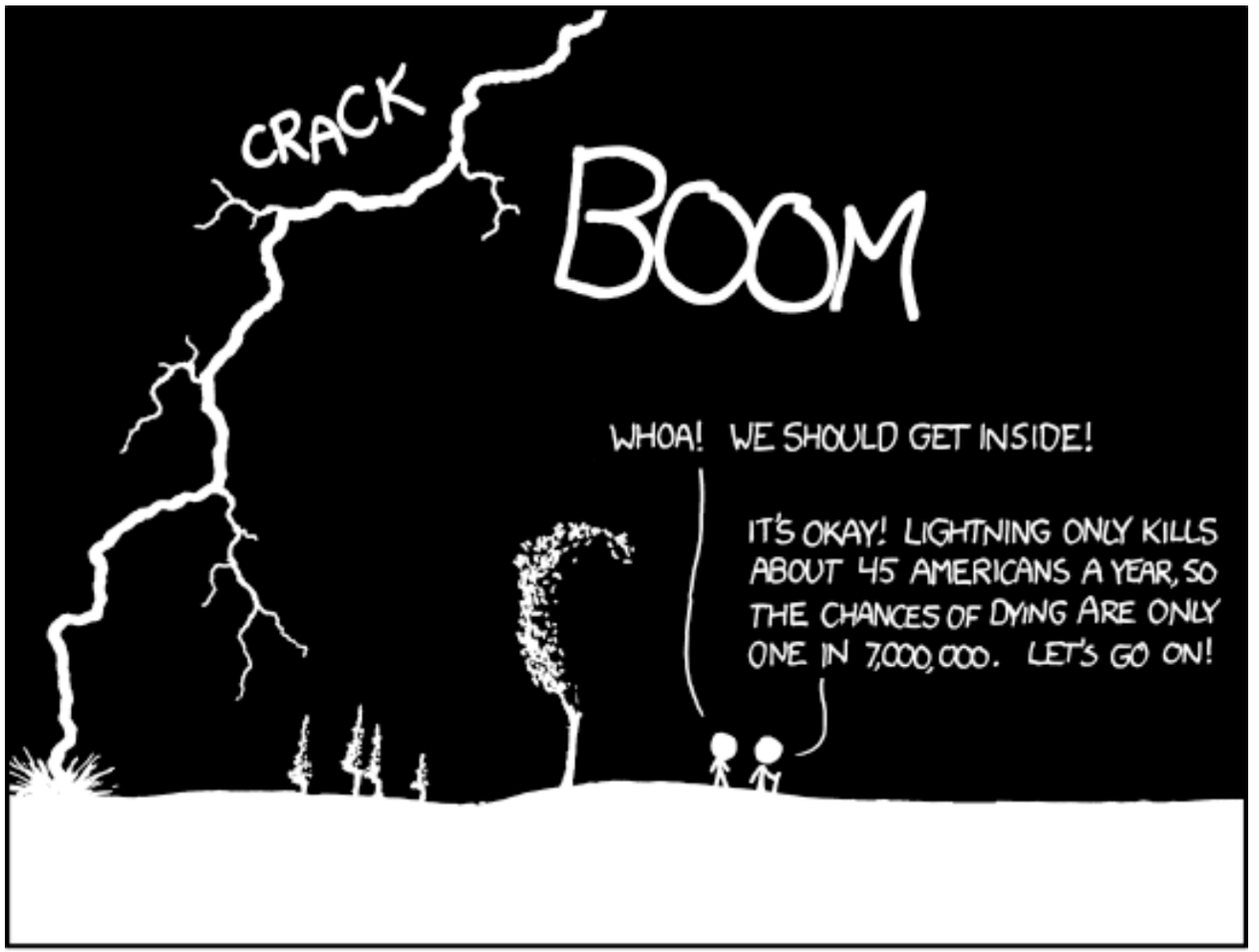
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Ans: $\Pr[2B \mid \geq 1B] = \dfrac{\Pr[2B \cap \geq 1B]}{\Pr[\geq 1B]} = \dfrac{\Pr[\{BB\}]}{\Pr[\{BB, BG, GB\}]} = \dfrac{1/4}{3/4} = \dfrac{1}{3}$

Importance of conditional probability



source: https://xkcd.com/795/

Bayes' rule

Rearranging the conditional probability formula, we get this equivalence:

$$\Pr[A \cap B] = \Pr[A \mid B] \Pr[B] = \Pr[B \mid A] \Pr[A]$$

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This is the simplest form of Bayes' rule. Often, the denominator is expanded further:

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The full form of Bayes' rule: $\Pr[A \mid B] = \dfrac{\Pr[B \mid A] \Pr[A]}{\Pr[B \mid A] \Pr[A] + \Pr[B \mid A'] \Pr[A']}$

But why? Bayes' rule lets you **flip a conditional**,
i.e. you have B given A but you want A given B

Bayes' rule exercise

Suppose a genetic condition occurs in roughly 1 out of 800 individuals (*prevalence*). A simple saliva test is available. If a person has the gene, the test is positive with 97% probability (*sensitivity*). If a person does not have the gene, a false positive occurs with 4% probability (*false positive rate*). A random person takes the test and gets positive, what's the probability they have the genetic condition?

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Hint: let $G =$ "have genetic condition", $P =$ "test is positive", we want $\Pr[G \mid P]$

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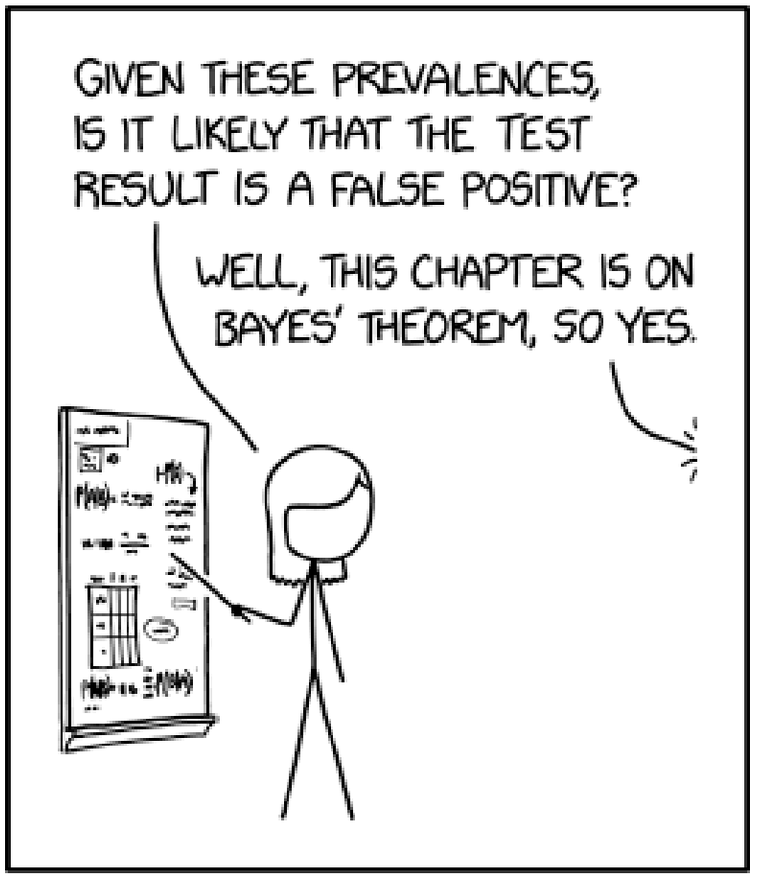
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Ans:

$$= \frac{(.97)(1/800)}{(.97)(1/800) + (.04)(799/800)} \approx 0.0295$$

A positive test has about 3% chance of being correct, due to the false positive rate being much higher than the rarity of the disease.

Bayes' rule



source: https://xkcd.com/2545/

Random variables, PMF/PDFs

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  - Example: binomial
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Note:

- PMFs output actual probabilities, PDFs output densities which are NOT probability (probabilities on PDFs are always areas of regions under the curve)
- PMFs and PDFs must satisfy all axioms of probability

RV examples

Discrete example: suppose X has the following PMF:

k	**0**	**1**	**2**	**3**	**4**
$\Pr[X = k]$	0.10	0.15	0.20	0.30	0.25

Exercises: 1. Check probability axioms are satisfied. 2. What's the probability of getting an even result? 3. Can you make a plot of this PMF?

RV examples

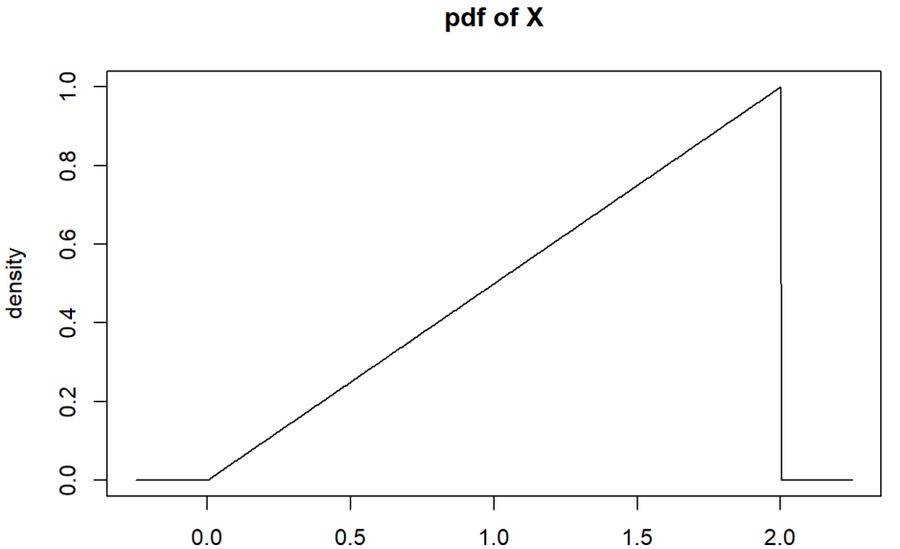
Discrete example: suppose X has the following PMF:

k	0	1	2	3	4
$\Pr[X = k]$	0.10	0.15	0.20	0.30	0.25

Exercises: 1. Check probability axioms are satisfied. 2. What's the probability of getting an even result? 3. Can you make a plot of this PMF?

Continuous example: suppose X outputs a real number in $[0, 2]$ with PDF $f(x) = x/2$.

Exercises: 1. Again check the probability axioms.
2. What's the probability of getting exactly 1?
How about the probability of getting > 0.5?
3. Again, can you plot this PDF yourself?



Cumulative distribution functions (CDFs)

For a RV X with a certain PMF/PDF, the **cumulative distribution function** (CDF) is defined

$$F_X(x) = \Pr[X \leq x]$$

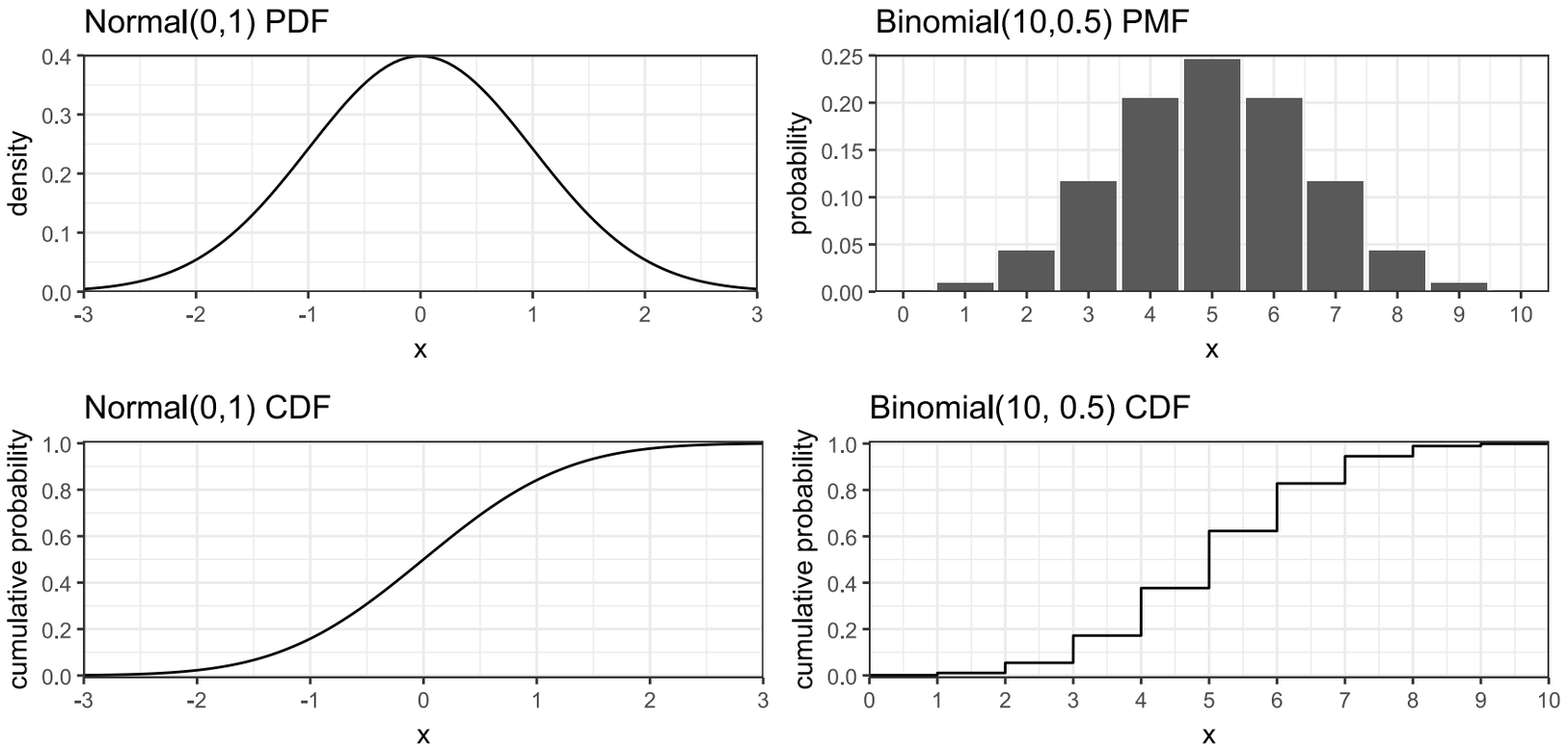
If $F_X(x) = p$, then x is the p-th **percentile** of X, i.e. the left-side integrated area under the PMF/PDF function from $-\infty$ to x equals p. (Note: X is the RV, x is the point!)

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CDF (cont.)

Any CDF $F(x)$ always satisfies these properties:

1. $F(-\infty) = 0, F(\infty) = 1$
2. F is **monotone non-decreasing** from left to right, i.e. if $a \leq b$, then $F(a) \leq F(b)$
3. a. If the RV is continuous, then F is continuous
   b. If the RV is discrete, then F has a "staircase" shape, with steps beginning at each possible outcome and having a height equal to that outcome's probability

CDF (cont.)

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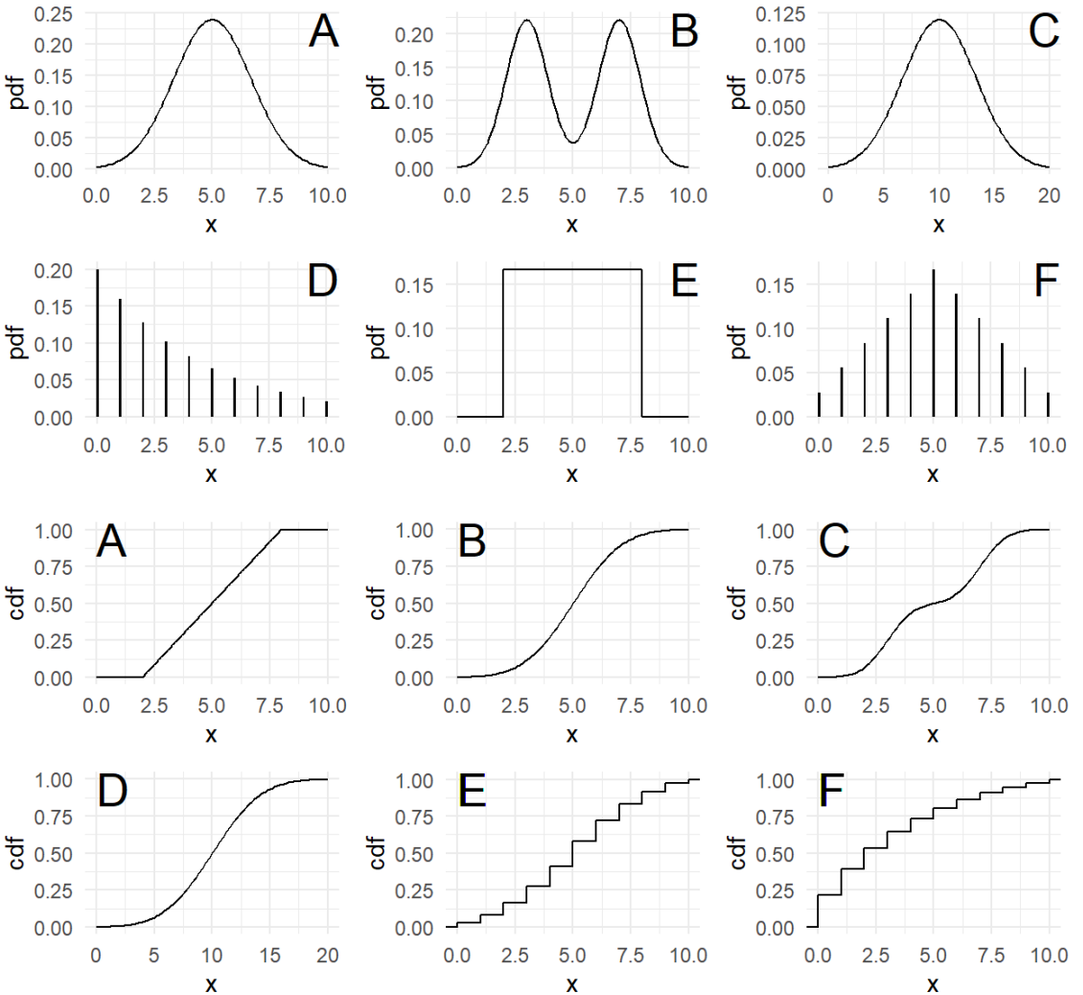
Additionally, note that the CDF can be used to calculate probabilities of events:

$$\Pr[a < X \leq b] = F(b) - F(a)$$

Also recall for discrete variables, $<$ vs $\leq$ and $>$ vs $\geq$ matters, but for continuous variables these are interchangeable, since probability of any single outcome $\Pr[X = x] = 0$

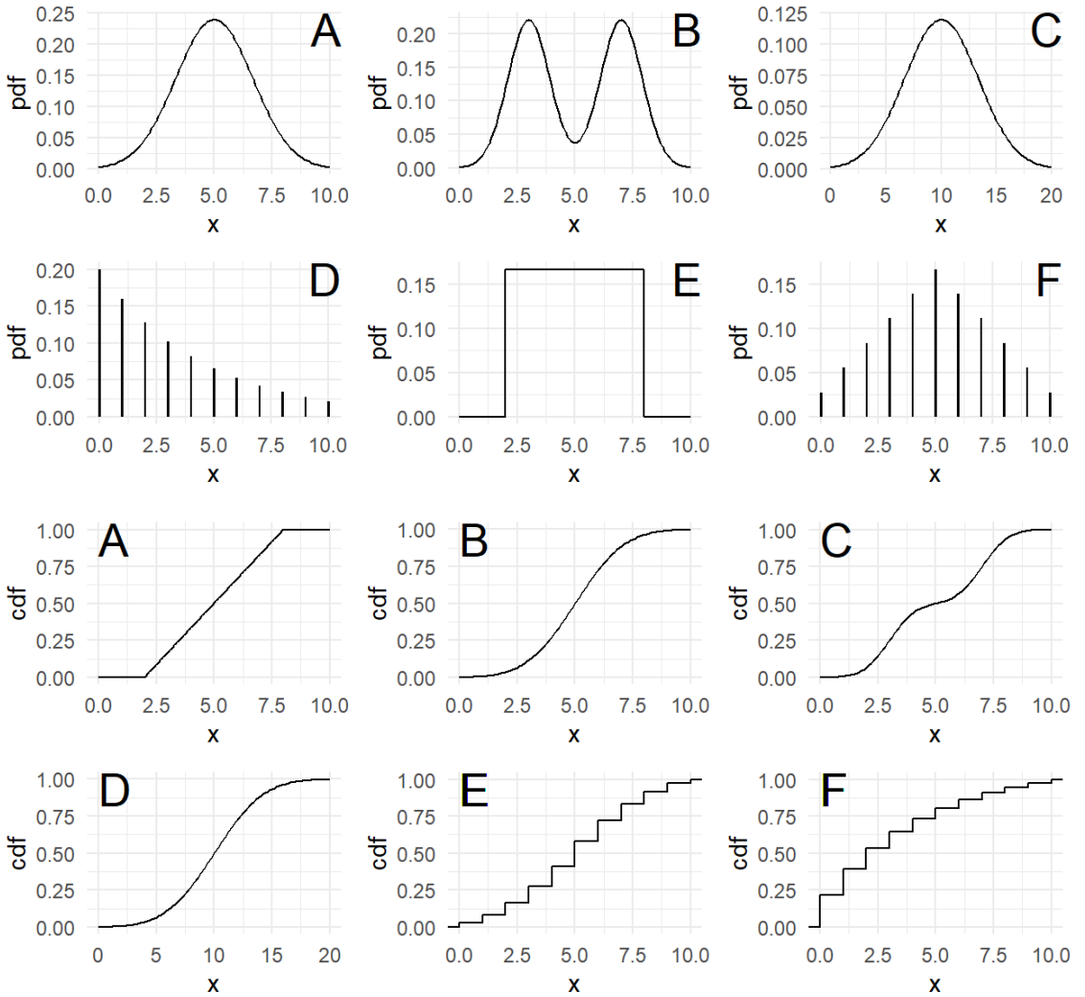
PMF/PDF, CDF exercise

Match the following densities to their corresponding CDFs:



PMF/PDF, CDF exercise

Match the following densities to their corresponding CDFs:



Densities A-F match in order: B, C, D, F, A, E

Inverse CDF (quantile)

The **inverse CDF** F^{-1}, also called the **quantile** function, inverts the CDF, i.e. does the "opposite". Given a probability p, it finds the **smallest** x such that $\Pr[X \leq x] \geq p$, i.e. it finds the smallest outcome x with at least p area to the left.

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Note inverse is the opposite function that "undoes" it, NOT the reciprocal.
Exercise: What are the inverses of: 1. $f(x) = x + 1$, 2. $f(x) = x^2$, 3. $f(x) = e^x$?

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- In R, PMF/PDFs are the `d...()` functions (d for density), e.g. `dnorm()`,
- CDFs are the `p...()` functions (p for percentile), e.g. `pnorm()`,
- and inverse CDFs are the `q...()` functions (q for quantile), e.g. `qnorm()`.

Shiny demo: https://www.statlab.wisc.edu/shiny/bwu62/340/pdf-cdf/ (*source*)

Expectation

The **expected value** (EV) $\mathrm{E}[X] = \sum_x x \cdot \Pr[X = x]$ or μ is the theoretical mean of X

In general $\mathrm{E}[f(X)] = \sum_x f(x) \cdot \Pr[X = x]$ gives the expectation of function f of X

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If $\mathrm{E}[X]$ exists, then the average of a large sample of independent observations will converge to it by the **Law of Large Numbers** (LLN). In other words, $\bar{x} \to \mathrm{E}[X]$ as $n \to \infty$

Exercise: show EV of fair n-sided die with $1, 2, \ldots, n$ on each side is $(n + 1)/2$

Expectation

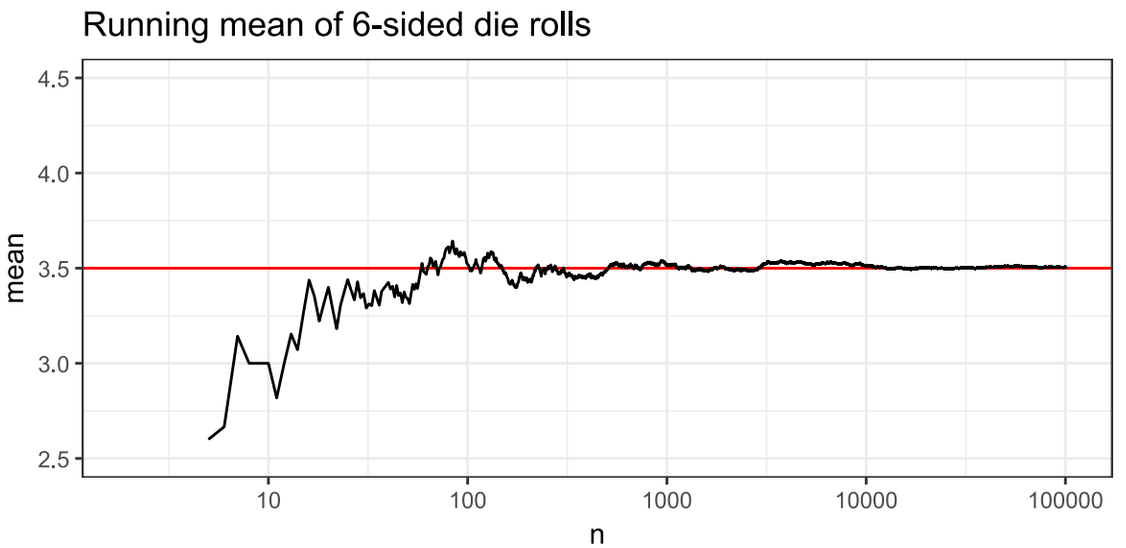
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Exercise: show EV of fair n-sided die with $1, 2, \ldots, n$ on each side is $(n+1)/2$

```r
# Example: show convergence of sample mean
options(scipen=1) # suppress scientific notation
set.seed(1) # set seed for repeatability
n = 1e5 # roll 6-sided die (EV=3.5) n times
rolls = sample(6, size=n, replace=TRUE)
m = cumsum(rolls)/seq_along(rolls)
tibble(x=2:n,y=m[-1]) %>% ggplot(aes(x,y)) +
  geom_hline(yintercept=3.5,color="red") +
  geom_line() + scale_x_log10() +
  scale_y_continuous(limits=c(2.5,4.5)) +
  theme_bw(base_size=18) + labs(x="n",y="mean",
  title="Running mean of 6-sided die rolls")
```



Variance

The **variance** $\operatorname{Var}[X] = \operatorname{E}[(X-\mu)^2] = \sum_x (x-\mu)^2 \cdot \Pr[X=x]$ or σ^2 is the mean squared distance of observations from μ. Its square root is called **standard deviation** or σ.

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Since this is also the average of a quantity, if it exists then by the LLN the sample variance $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$ converges to it, i.e. $S^2 \to \sigma^2$ as $n \to \infty$.

Exercise: show variance of fair 6-sided die $= 35/12 \approx 2.92$

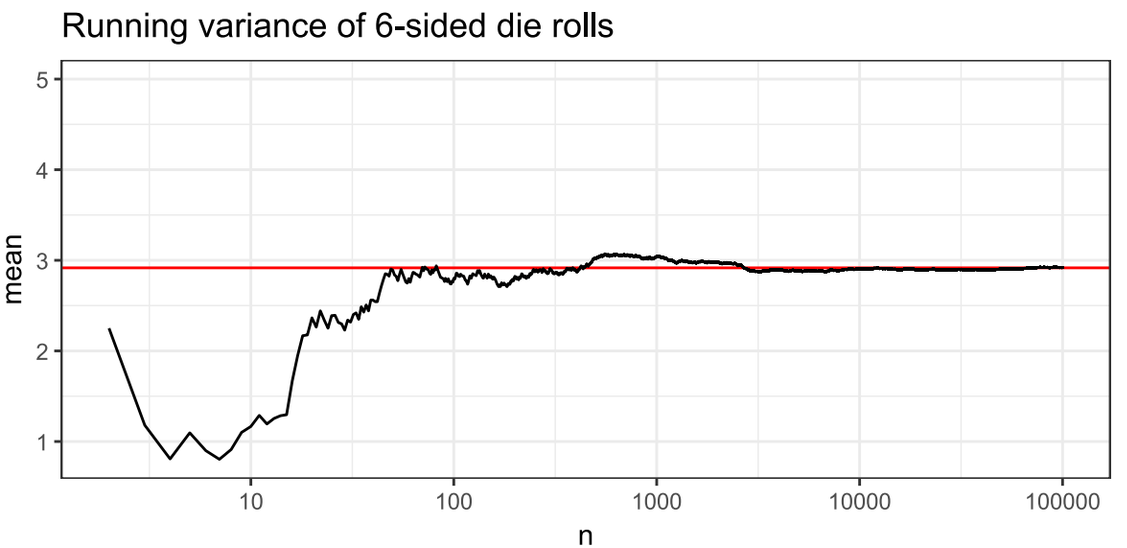
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Exercise: show variance of fair 6-sided die $= 35/12 \approx 2.92$

```r
# Example: show convergence of sample variance
set.seed(3) # set seed for repeatability
n = 1e5 # roll 6-sided die n times
rolls = sample(6, size=n, replace=TRUE)
m = cumsum(rolls)/seq_along(rolls)
v = cumsum((rolls-m)^2)/(seq_along(rolls)-1)
tibble(x=2:n,y=v[-1]) %>% ggplot(aes(x,y)) +
  geom_hline(yintercept=35/12,color="red") +
  geom_line() + scale_x_log10() +
  scale_y_continuous(limits=c(.8,5)) +
  theme_bw(base_size=18) + labs(x="n",y="mean",
  title="Running variance of 6-sided die rolls")
```



Continuous E, Var

For continuous X, replace PMF and summation with PDF $f(x)$ and integral

$$\mathrm{E}[X] = \int_{\Omega} x f(x)\,\mathrm{dx} \qquad \mathrm{Var}[X] = \int_{\Omega} (x-\mu)^2 f(x)\,\mathrm{dx}$$

EX: show for previous example PDF $f(x) = x/2$ over $[0, 2]$ that $\mu = 4/3, \sigma^2 = 2/9$

Continuous E, Var

For continuous X, replace PMF and summation with PDF $f(x)$ and integral

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EX: show for previous example PDF $f(x) = x/2$ over $[0, 2]$ that $\mu = 4/3, \sigma^2 = 2/9$

$$\mathrm{E}[X] = \int_0^2 x\left(\frac{x}{2}\right)\,\mathrm{dx} = \frac{1}{6}x^3\Big|_0^2 = \frac{4}{3}$$

$$\begin{aligned}\mathrm{Var}[X] &= \int_0^2 \left(x-\frac{4}{3}\right)^2\left(\frac{x}{2}\right)\mathrm{dx} \\ &= \frac{1}{2}\int_0^2 \left[x^3 - \frac{8}{3}x^2 + \frac{16}{9}x\right]\mathrm{dx} \\ &= \frac{1}{2}\left|\frac{1}{4}x^4 - \frac{8}{9}x^3 + \frac{8}{9}x^2\right|_0^2 = \frac{2}{9}\end{aligned}$$

RVs covered in 340

- Discrete:
  - Binomial
    - Special case for $n = 1$ called Bernoulli
  - Geometric
  - Poisson
  - Discrete uniform
- Continuous:
  - Normal
  - (Continuous) uniform
  - Exponential

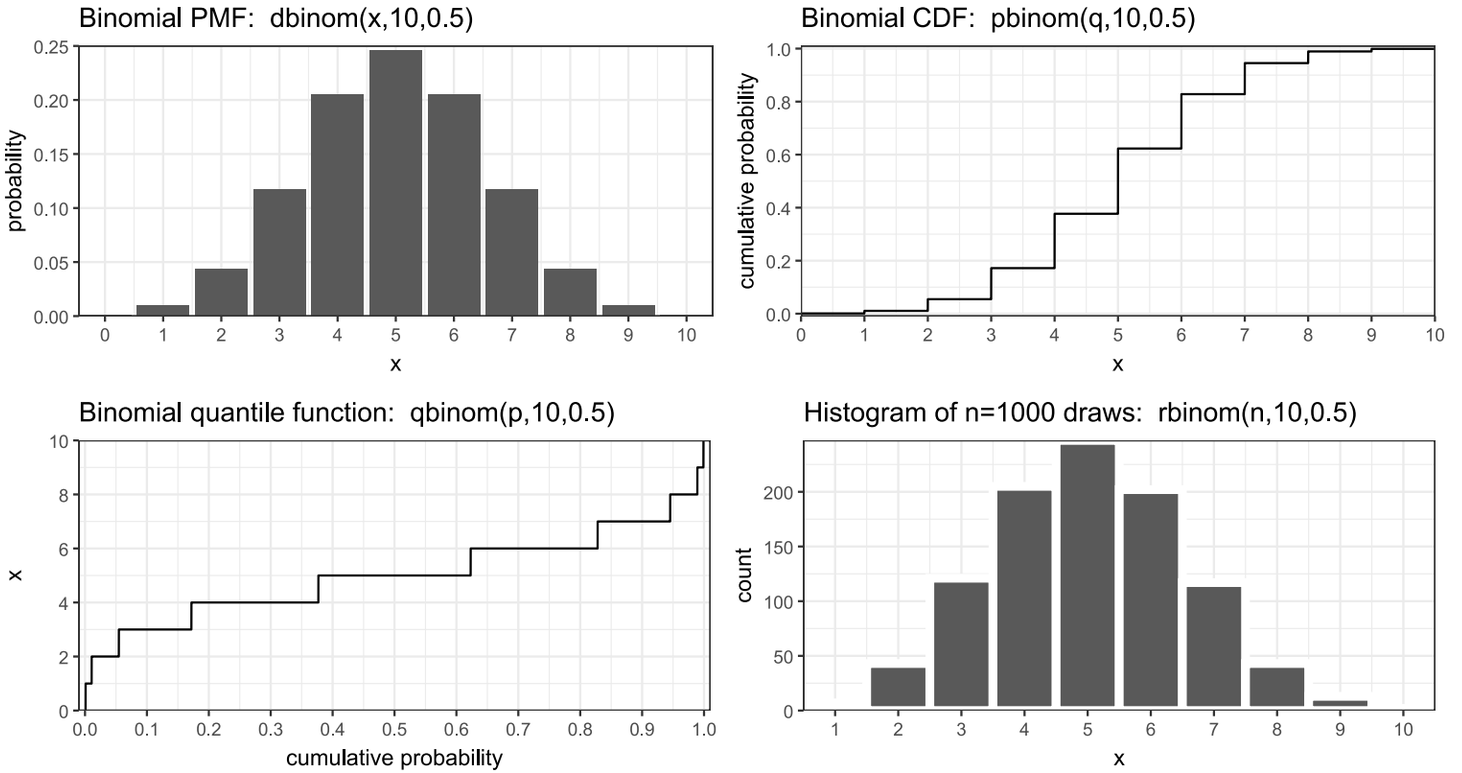
Binomial(n, p)

- Models **number of successes** in n independent trials, each with chance p of success
- Examples: number of heads in flips, number of patients responding to treatment, etc.
- Outcomes: 0, 1, ..., n. Moments: $\mu = np,\ \sigma^2 = np(1-p)$
- PMF: $\Pr[X = k] = \binom{n}{k} p^k (1-p)^{n-k}$

- $n = 1$ case called Bernoulli(p)

Example:

Binomial$(10, 0.5)$
$\mu = 5,\ \sigma \approx 1.58$



Geometric(p)

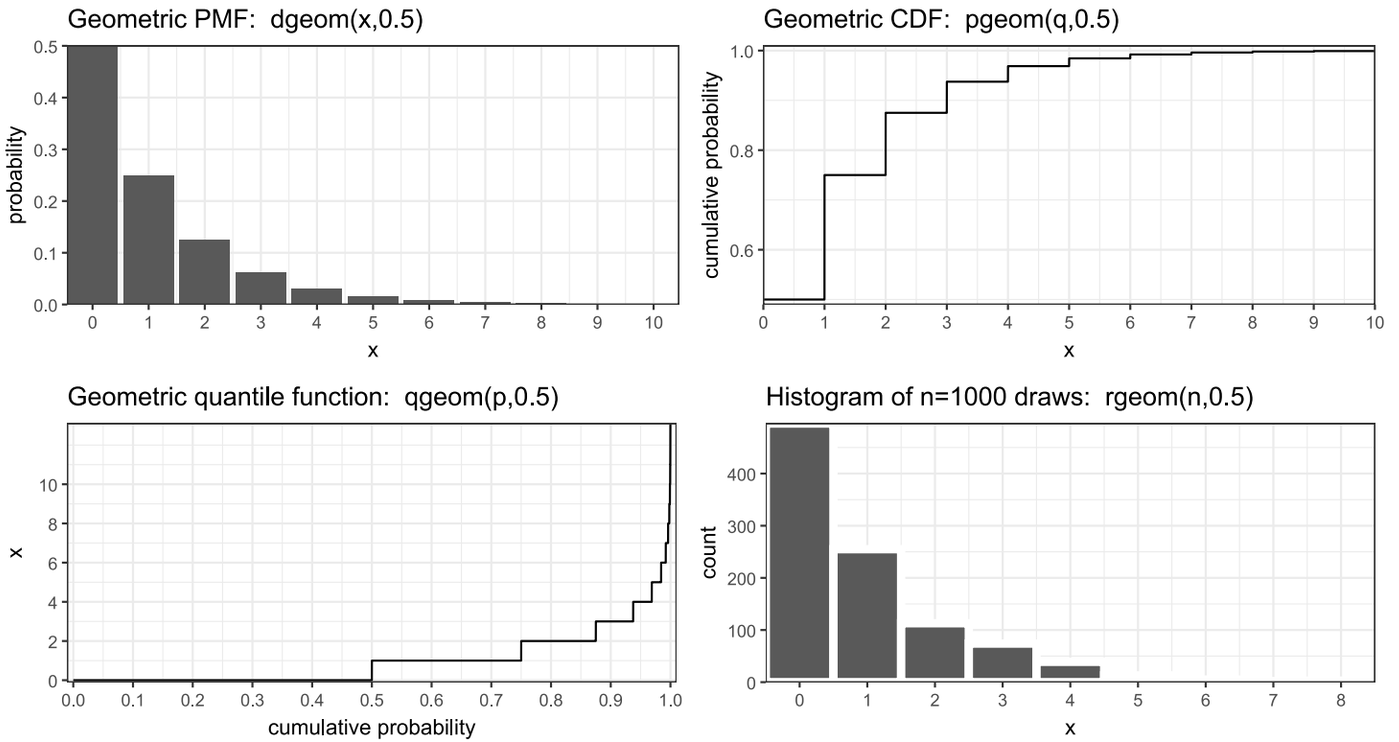
- Models **number of trials *before 1st success***, each trial independently chance p of success
- Examples: flipping coin until we get heads, number of tries before making a shot, etc.
- Outcomes: 0, 1, 2, … . Moments: $\mu = (1-p)/p,\ \sigma^2 = (1-p)/p^2$
- PMF: $\Pr[X = k] = p(1-p)^k$, CDF: $\Pr[X \le k] = 1 - (1-p)^{\lfloor k \rfloor + 1}$

- Note: there's a competing definition of "number of trials ***until*** first success, instead of ***before***. We will prefer the ***before*** version!

Example:

Geometric(0.5)

$\mu = 1,\ \sigma \approx 1.44$

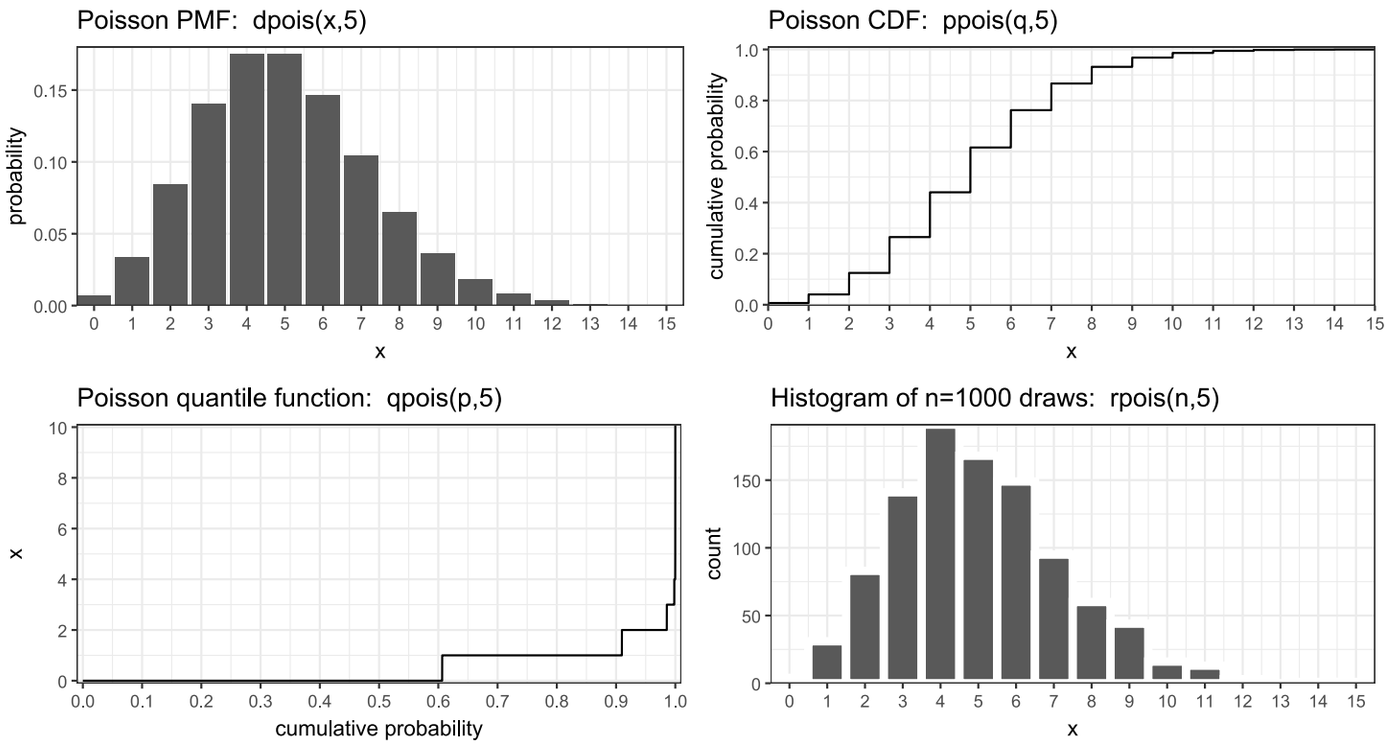


Poisson(λ)

- Models **count of occurrences** when events happen independently with average rate λ
- Examples: number of fish caught on trip, number of weeds in certain area, etc.
- Outcomes: 0, 1, 2, … . Moments: $\mu = \sigma^2 = \lambda$
- PMF: $\Pr[X = k] = \lambda^k e^{-\lambda}/k!$

- λ usually defined over some amount of time/ space/etc & scales with it
- Can be derived from limit of binomial, see: https://mathcenter.oxford.emory.edu/site/math117/connectingPoissonAndBinomial

Example: Poisson(5),
i.e. something happening on average 5 times per ___

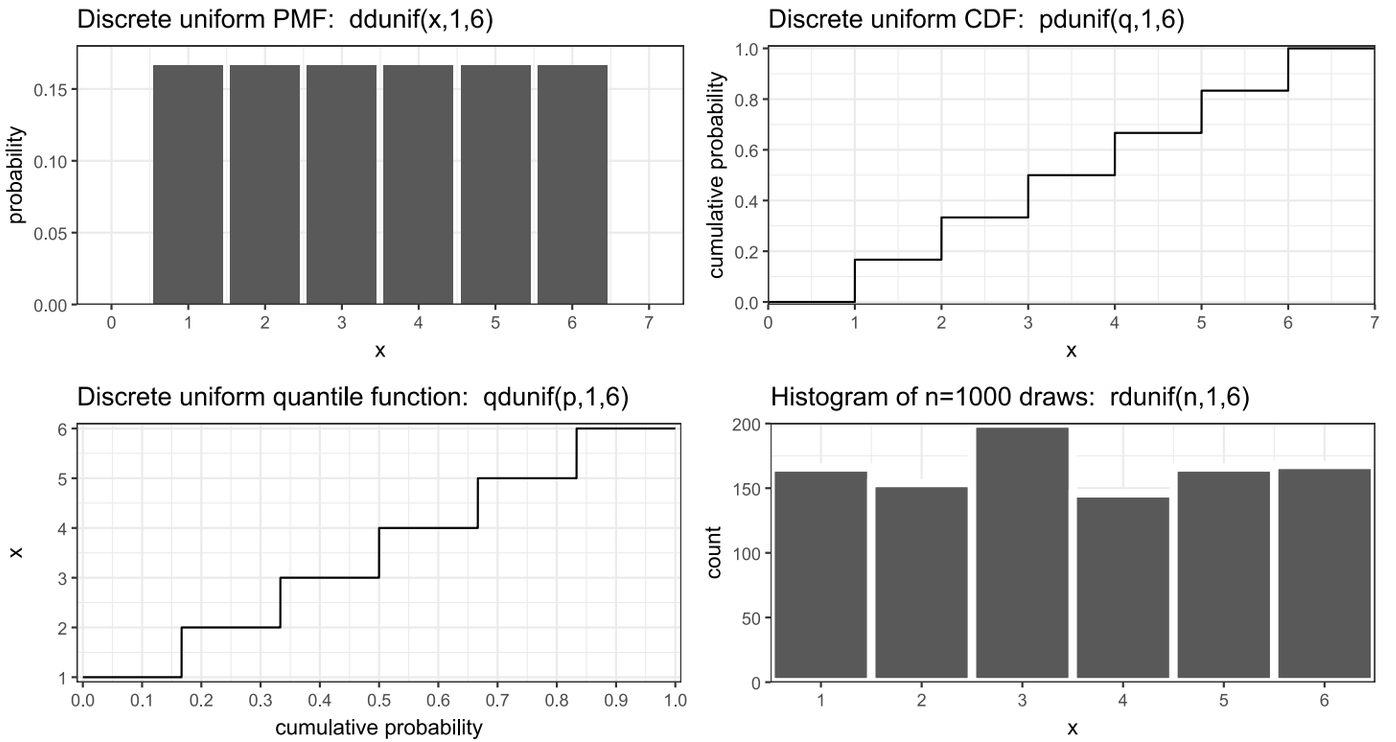


Discrete uniform(a, b)

- Models n-sided fair die with integers $a, a+1, \ldots, b$ on it (note $n = b - a + 1$)
- Other examples: lottery draws, seconds value when you check the time, etc.
- Outcomes: $a, a+1, ..., b$. Moments: $\mu = (a+b)/2,\ \sigma^2 = (n^2-1)/12$
- PMF: $\Pr[X = k] = 1/n$

Example:
Discrete uniform$(1, 6)$,
i.e. fair 6-sided die roll
(using `extraDistr` package)

$\mu = 3.5,\ \sigma \approx 1.71$

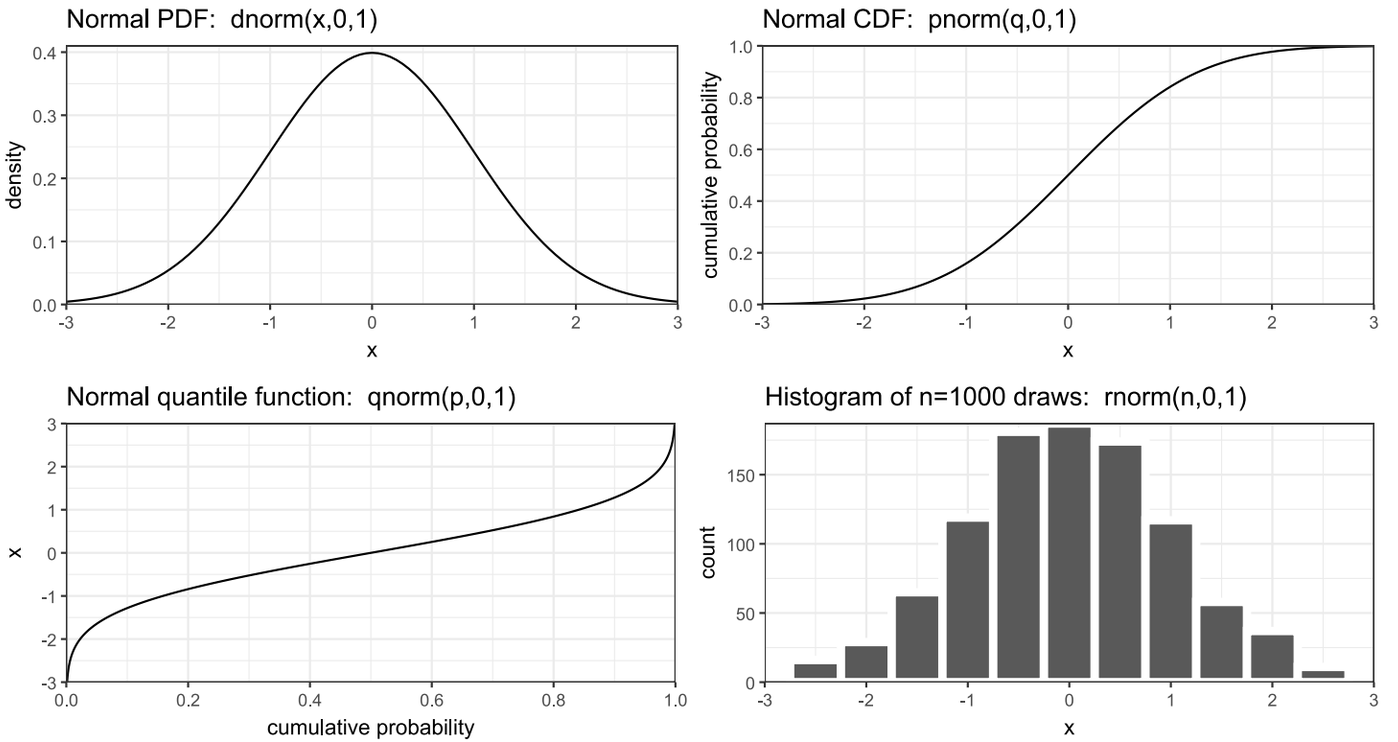


Normal(μ, σ^2)

- Most things that are the sum or mean of many smaller independent components are well modeled by the normal (due to the Central Limit Theorem)
- Outcomes: $(-\infty, \infty)$. Moments: $\mu = \mu,\ \sigma^2 = \sigma^2$
- PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

- Note: in 340, we will prefer the parametrization Normal(μ, σ^2) instead of Normal(μ, σ) from 240
- Remember empirical rule: 68/95/99.7% of data within ±1/2/3 SDs

Example:
Normal$(0, 1)$, i.e. the standard normal Z

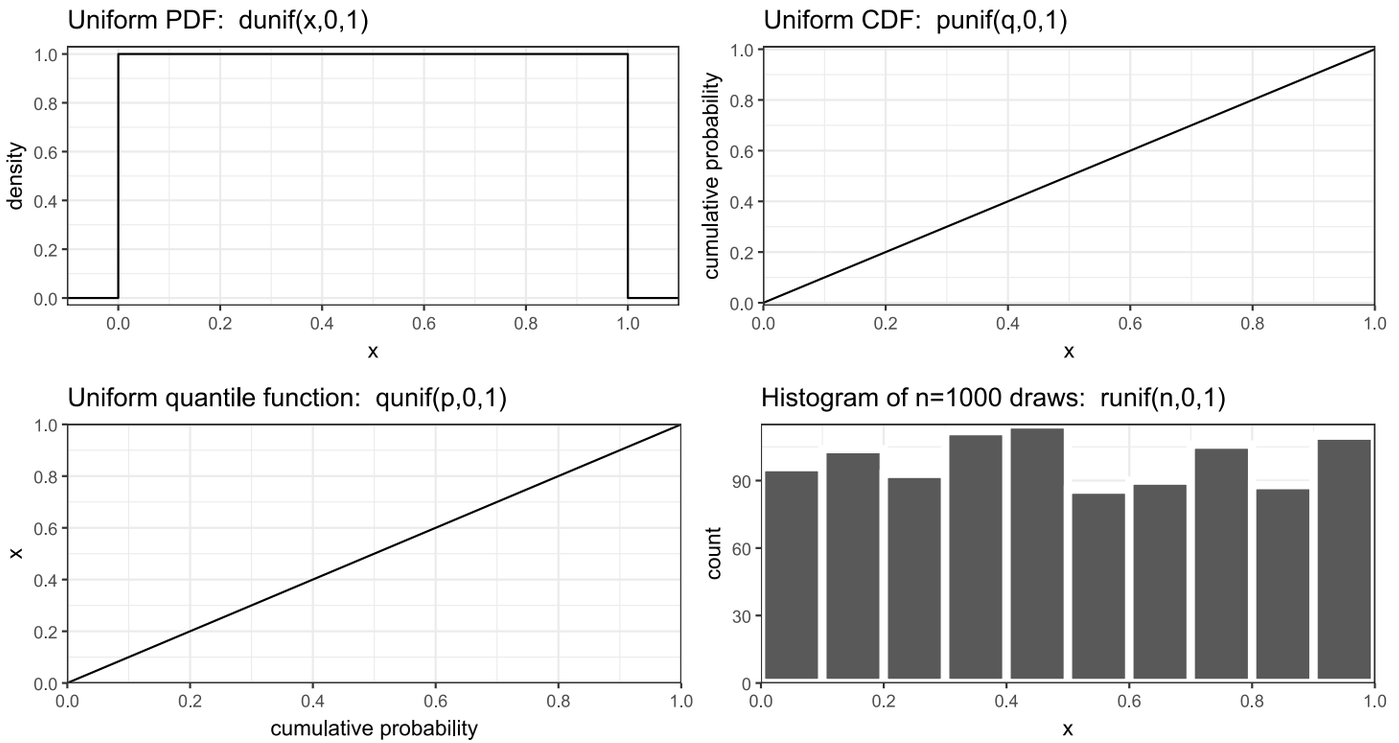


(Continuous) Uniform(a, b)

- Continuous version of discrete uniform, models when anything in $[a, b]$ equally likely
- Examples: arbitrary angles, percentiles (more on this later), etc.
- Outcomes: $[a, b]$. Moments: $\mu = (a+b)/2,\ \sigma^2 = (b-a)^2/12$
- PDF: $f(x) = 1/(b-a)$, CDF: $F(x) = (x-a)/(b-a)$

Example: Uniform$(0, 1)$

$\mu = 0.5,\ \sigma^2 = 1/12$

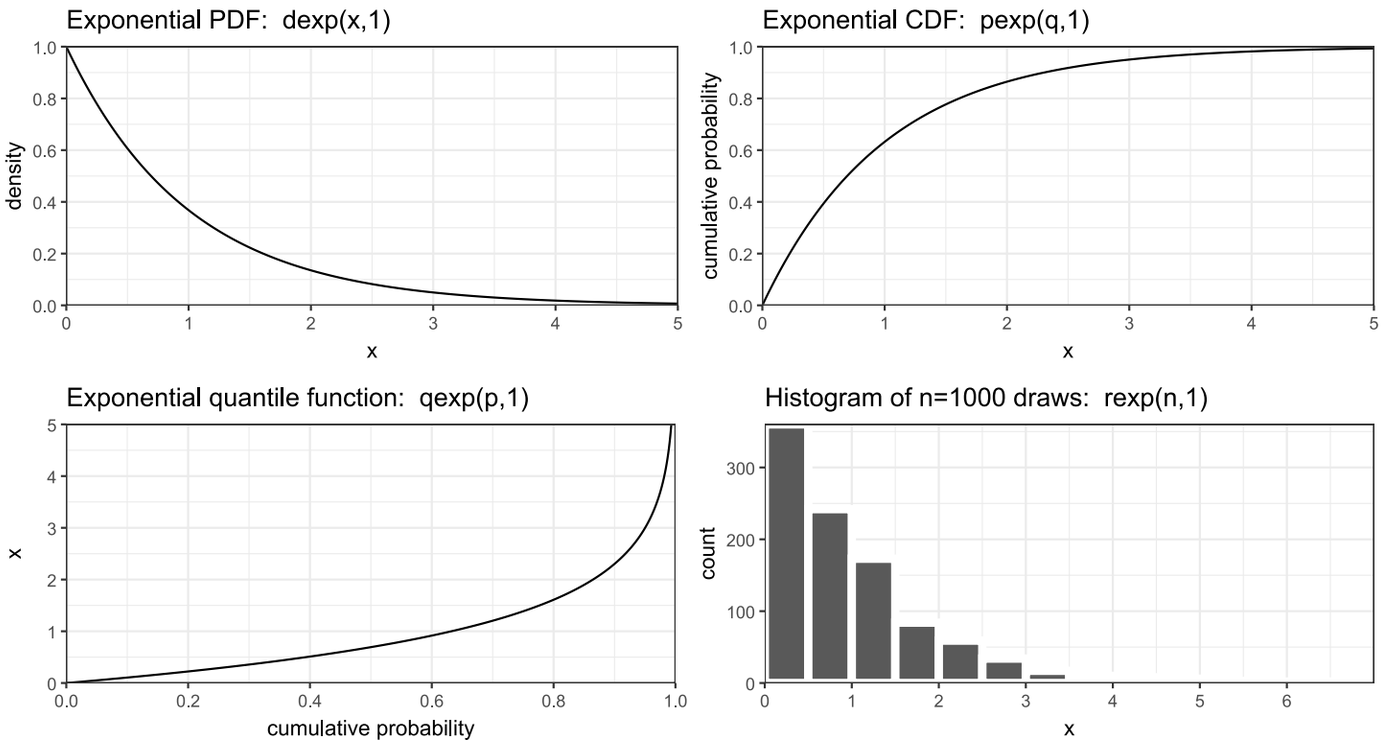


Exponential(λ)

- Continuous version of geometric, models **wait time between Poisson(λ) events**
- Examples: time between earthquakes, time between calls at call center, etc.
- Outcomes: $[0, \infty)$. Moments: $\mu = 1/\lambda,\ \sigma^2 = 1/\lambda^2$
- PDF: $f(x) = \lambda e^{-\lambda x}$, CDF: $F(x) = 1 - e^{-\lambda x}$

Example: Exponential(1)

$\mu = 1,\ \sigma^2 = 1$



RV exercises

1. Which RVs always have P(X=0) > 0? Choose ALL that apply!
   a. Normal
   b. Uniform
   c. Exponential
   d. Poisson
   e. Geometric

2. Which RVs have the largest mean? If multiple are tied, choose ALL that apply!
   a. Binomial(20,0.2)
   b. Normal(4,4^2)
   c. Geometric(0.2)
   d. Poisson(4)
   e. Exponential(0.25)

3. For each, choose ONE RV as the BEST model, using each RV no more than once! RVs: Bernoulli, Binomial, Geometric, Poisson, Normal, Discrete uniform, Exponential.
   a. Number of days since the last injury at a factory.
   b. Number of students in a class who fail.
   c. Scores of students on a standardized exam.
   d. Rank of card drawn from shuffled standard (52-card) deck.
   e. Whether or not your classmate John Smith gets an A.

RV exercises

1. Which RVs always have P(X=0) > 0? Choose ALL that apply!
   a. Normal
   b. Uniform
   c. Exponential
   d. **Poisson** $\Pr[0] = e^{-\lambda}$
   e. **Geometric** $\Pr[0] = p$

2. Which RVs have the largest mean? If multiple are tied, choose ALL that apply!
   a. **Binomial(20,0.2)**
   b. **Normal(4,4^2)**
   c. **Geometric(0.2)**
   d. **Poisson(4)**
   e. **Exponential(0.25)** (all have mean 4)

3. For each, choose ONE RV as the BEST model, using each RV no more than once! RVs: Bernoulli, Binomial, Geometric, Poisson, Normal, Discrete uniform, Exponential.
   a. Number of days since the last injury at a factory: **Geometric**
   b. Number of students in a class who fail: **Binomial** (if μ small, poisson may work too)
   c. Scores of students on a standardized exam: **Normal**
   d. Rank of card drawn from shuffled standard (52-card) deck: **Discrete uniform**
   e. Whether or not your classmate John Smith gets an A: **Bernoulli**

Independence of RVs

Like events, RVs can also be dependent or independent. For RVs X, Y, they are independent if **any two events** related to them are also independent.

I.e. if for any two events $E_X \subseteq \Omega_X, E_Y \subseteq \Omega_Y$ relating to X, Y it's true that:

$$\Pr[X \in E_X, Y \in E_Y] = \Pr[X \in E_X] \Pr[Y \in E_Y]$$

Then X, Y are independent. Otherwise, they're dependent.

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Then X, Y are independent. Otherwise, they're dependent.

We often assume observations in samples are **independent & identically distributed** (iid). E.g. suppose n students take an exam with scores normalized to have mean 100 and sd 15. We might say: let $X_1, X_2, \dots, X_n$ be students' exam scores where

$$X_i \overset{\text{iid}}{\sim} \text{Normal}(100, 15^2)$$

Then for any two students X_i, X_j and any two events E_i, E_j, we have

$$\Pr[X_i \in E_i, X_j \in E_j] = \Pr[X_i \in E_i] \Pr[X_j \in E_j]$$

Expectation properties

It's easy to show E is linear. Let X be a discrete RV and a, b be constants. Then:

$$\begin{aligned}
\mathrm{E}[aX + b] &= \sum_x (ax + b) \Pr[X = x] \\
&= a \sum_x x \Pr[X = x] + b \sum_x \Pr[X = x] \\
&= a \, \mathrm{E}[X] + b
\end{aligned}$$

Expectation properties

It's easy to show E is linear. Let X be a discrete RV and a, b be constants. Then:

$$\begin{aligned}\mathrm{E}[aX+b] &= \sum_x (ax+b)\Pr[X=x] \\ &= a\sum_x x\Pr[X=x] + b\sum_x \Pr[X=x] \\ &= a\,\mathrm{E}[X] + b\end{aligned}$$

Replacing summations with integrals completes the proof for continuous RVs. Similarly you can show for both discrete and continuous RVs for any RVs X, Y that:

$$\mathrm{E}[X+Y] = \mathrm{E}[X] + \mathrm{E}[Y]$$

These are the defining properties of a **linear function**

Variance properties

Var is a bit odd. Let X be an RV (discrete or continuous) and a, b be constants. Then:

$$\begin{aligned}\operatorname{Var}[aX+b] &= \operatorname{E}[(aX+b-\operatorname{E}[aX+b])^2] \\ &= \operatorname{E}[(aX-a\operatorname{E}[X])^2] \\ &= a^2\operatorname{E}[(X-\mu)^2] \;=\; a^2\operatorname{Var}[X]\end{aligned}$$

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How about sums of variables? Let X, Y be RVs, then:

$$\begin{aligned}\operatorname{Var}[X+Y] &= \operatorname{E}[(X+Y-\operatorname{E}[X+Y])^2] \\ &= \operatorname{E}[((X-\operatorname{E}[X])+(Y-\operatorname{E}[Y]))^2] \\ &= \operatorname{E}[(X-\operatorname{E}[X])^2+(Y-\operatorname{E}[Y])^2+2(X-\operatorname{E}[X])(Y-\operatorname{E}[Y])] \\ &= \operatorname{Var}[X]+\operatorname{Var}[Y]+2\operatorname{Cov}(X,Y)\end{aligned}$$

Variance properties

Var is a bit odd. Let X be an RV (discrete or continuous) and a, b be constants. Then:

$$\begin{aligned}\operatorname{Var}[aX+b] &= \operatorname{E}[(aX+b-\operatorname{E}[aX+b])^2] \\ &= \operatorname{E}[(aX-a\operatorname{E}[X])^2] \\ &= a^2\operatorname{E}[(X-\mu)^2] \;=\; a^2\operatorname{Var}[X]\end{aligned}$$

How about sums of variables? Let X, Y be RVs, then:

$$\begin{aligned}\operatorname{Var}[X+Y] &= \operatorname{E}[(X+Y-\operatorname{E}[X+Y])^2] \\ &= \operatorname{E}[((X-\operatorname{E}[X])+(Y-\operatorname{E}[Y]))^2] \\ &= \operatorname{E}[(X-\operatorname{E}[X])^2+(Y-\operatorname{E}[Y])^2+2(X-\operatorname{E}[X])(Y-\operatorname{E}[Y])] \\ &= \operatorname{Var}[X]+\operatorname{Var}[Y]+2\operatorname{Cov}(X,Y)\end{aligned}$$

Where the **covariance** of X, Y, i.e. $\operatorname{Cov}(X,Y)=\operatorname{E}[(X-\operatorname{E}[X])(Y-\operatorname{E}[Y])]$ describes how much X, Y change together or apart.

If X, Y are independent, then $\operatorname{Cov}(X,Y)=0$ and $\operatorname{Var}[X+Y]=\operatorname{Var}[X]+\operatorname{Var}[Y]$

Covariance, correlation, independence

The covariance is actually related to the correlation coefficient $\rho_{X,Y}$, which is defined:

$$\rho_{X,Y} = \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}[X]\operatorname{Var}[Y]}}$$

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Ex: let X be discrete uniform$(-1, 1)$, let $Y = X^2$. Show X, Y uncorrelated but dependent

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Ans: $\operatorname{Cov}(X,Y) = \operatorname{E}[(X - \operatorname{E}[X])(Y - \operatorname{E}[Y])] = \operatorname{E}[X(X^2 - \frac{2}{3})] = \cancel{\operatorname{E}[X^3]} - \frac{2}{3}\cancel{\operatorname{E}[X]} = 0$, but $\Pr[X=1, Y=1] = \frac{1}{3}$ whereas $\Pr[X=1] \cdot \Pr[Y=1] = \frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9}$, so uncorrelated but dependent

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Additional exercises for home: (*hint: remember μ, σ^2 are always constants for any RV!*)

1. Show also $\operatorname{Cov}(X,Y) = \operatorname{E}[XY] - \operatorname{E}[X]\operatorname{E}[Y]$
2. Show also $\operatorname{Var}[X] = \operatorname{E}[X^2] - (\operatorname{E}[X])^2$

Summary of identities

$$
\begin{aligned}
\mathrm{E}[aX+b] &= a\,\mathrm{E}[X]+b \\
\mathrm{E}[X+Y] &= \mathrm{E}[X]+\mathrm{E}[Y] \\
\mathrm{Var}[X] &= \mathrm{E}[X^2]-(\mathrm{E}[X])^2 \\
\mathrm{Var}[aX+b] &= a^2\,\mathrm{Var}[X] \\
\mathrm{Var}[X+Y] &= \mathrm{Var}[X]+\mathrm{Var}[Y]+2\,\mathrm{Cov}(X,Y) \\
\mathrm{Cov}(X,Y) &= \mathrm{E}[(X-\mathrm{E}[X])(Y-\mathrm{E}[Y])] \\
&= \mathrm{E}[XY]-\mathrm{E}[X]\,\mathrm{E}[Y] = \rho_{X,Y}\cdot\sigma_X\cdot\sigma_Y
\end{aligned}
$$

If X, Y independent, then additionally:

$$
\begin{aligned}
\mathrm{Cov}(X,Y) &= 0 \\
\mathrm{Var}[X+Y] &= \mathrm{Var}[X]+\mathrm{Var}[Y] \\
\mathrm{E}[XY] &= \mathrm{E}[X]\,\mathrm{E}[Y]
\end{aligned}
$$

Central limit theorem (CLT)

The CLT crucially explains why the normal distribution is so useful. The common version:

Let $X_1, \ldots, X_n$ be iid with $\mathrm{E}[X_i] = \mu,\ \mathrm{Var}[X_i] = \sigma^2 < \infty$, then as $n \to \infty$,

$$\bar{X} \sim \mathrm{Normal}(\mu, \tfrac{\sigma^2}{n})$$

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The normality part is nontrivial to prove and beyond our scope, but the expectation and variance are relatively easy for us to show:

$$\begin{aligned}
\mathrm{E}[\bar{X}] &= \mathrm{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] \\
&= \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}[X_i] \\
&= \frac{1}{n}\sum_{i=1}^{n} \mu \\
&= \frac{1}{n}(n\mu) \ = \ \mu
\end{aligned}$$

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&= \frac{1}{n}(n\mu) \;=\; \mu
\end{aligned}$$

$$\begin{aligned}
\mathrm{Var}[\bar{X}] &= \mathrm{Var}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] \\
&= \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}[X_i] \quad (\text{since } X_i \text{ iid}) \\
&= \frac{1}{n^2}\sum_{i=1}^{n} \sigma^2 \\
&= \frac{1}{n^2}(n\sigma^2) \;=\; \frac{\sigma^2}{n}
\end{aligned}$$

Bonus: Bessel's correction

We can now actually prove that $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$ makes sense. Suppose X_i are iid with $\mathrm{E}[X_i] = \mu$, and $\mathrm{Var}[X_i] = \sigma^2$. Also recall $\mathrm{Var}[X] = \mathrm{E}[X^2] - (\mathrm{E}[X])^2$. Then:

$$
\begin{aligned}
\mathrm{E}\left[\tfrac{1}{n-1}\sum(X_i - \bar{X})^2\right] &= \tfrac{1}{n-1}\,\mathrm{E}\left[\sum(X_i^2 + \bar{X}^2 - 2X_i\bar{X})\right] \\
&= \tfrac{1}{n-1}\left(\sum \mathrm{E}[X_i^2] + \sum \mathrm{E}[\bar{X}^2] - 2\,\mathrm{E}\left[\bar{X}\cdot\sum X_i\right]\right) \\
&= \tfrac{1}{n-1}\left(n\,\mathrm{E}[X_i^2] + \cancel{n\,\mathrm{E}[\bar{X}^2]} - \cancel{2}\,n\,\mathrm{E}[\bar{X}^2]\right) \\
&= \tfrac{1}{n-1}\left(n\Big(\mathrm{Var}[X_i] + (\mathrm{E}[X_i])^2\Big) - n\Big(\mathrm{Var}[\bar{X}] + (\mathrm{E}[\bar{X}])^2\Big)\right) \\
&= \tfrac{1}{n-1}\left(n(\sigma^2 + \mu^2) - n(\tfrac{\sigma^2}{n} + \mu^2)\right) \\
&= \tfrac{1}{n-1}(n\sigma^2 - \sigma^2) \\
&= \sigma^2
\end{aligned}
$$

Since $\mathrm{E}[S^2] = \sigma^2$, this estimator is **unbiased** (more on this in later chapters).

Bonus: E/Var for continuous RV

For continuous RV X with PDF $f(x)$, E/Var defined as:

$$\mathrm{E}[X] = \int_\Omega x f(x) dx \qquad \mathrm{Var}[X] = \int_\Omega (x-\mu)^2 f(x) dx$$

Example: let $X = \mathrm{Uniform}(a, b)$, then $f(x) = \frac{1}{b-a}$, and

$$\begin{aligned}
\mathrm{E}[X] &= \int_a^b x \, \frac{1}{b-a} \, dx \\
&= \frac{1}{2}\left[\frac{x^2}{b-a}\right]_a^b \\
&= \frac{1}{2}\frac{b^2-a^2}{b-a} \\
&= \frac{a+b}{2}
\end{aligned}$$

$$\begin{aligned}
\mathrm{Var}[\bar{X}] &= \int_a^b \left(x - \frac{1}{2}(a+b)\right)^2 \frac{1}{b-a} \, dx \\
&= \frac{1}{b-a}\left[\frac{1}{3}\left(x - \frac{a+b}{2}\right)^3\right]_a^b \\
&= \frac{1}{24(b-a)} \cdot \left[(2x-a-b)^3\right]_a^b \\
&= \frac{1}{24(b-a)} \cdot \left[2(b-a)^3\right] \;=\; \frac{(b-a)^2}{12}
\end{aligned}$$

Additional readings/practice

Some parts of the notes if you want more readings/practice:

- Additional examples of each RV in section 6.2 on RV families and section 7
- 6.3 on approximating variables
- Try some practice exercises in section 5 and section 8